Learning a Margin of Normalcy for Zero-Day Network Intrusion Detection: Masked Autoencoders and One-Class SVMs on NSL-KDD Flow Features

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March 22, 2026

Abstract

Signature-based intrusion detection systems fail silently against *zero-day* attacks for which no templates exist. We frame enterprise and industrial monitoring as *semi-supervised anomaly detection*: models observe only benign connection traffic during training and score deviations at deployment. On the NSL-KDD benchmark, we combine (i) skew-driven logarithmic transforms with scalable one-hot encoding, (ii) a tabular masked autoencoder trained with a BERT-style masked reconstruction objective, (iii) RBF one-class SVMs in both raw feature space and learned latent space, and (iv) a calibrated hybrid score. Using the standard NSL train/test protocol with a stratified validation split for threshold selection, we obtain test ROC-AUC of 0.942 for reconstruction-based detection and 0.920 for one-class SVM on raw features; a naïve linear fusion of reconstruction error and latent SVM margin *degrades* ranking quality (ROC-AUC 0.781), which we analyse as a failure mode of score aggregation under heterogeneous scales. These results support reconstruction and kernel methods as strong baselines while motivating more principled fusion for hybrid designs.

Keywords: anomaly detection, NSL-KDD, masked autoencoder, one-class SVM, semi-supervised learning, cyber-physical security.

1 Introduction

The earliest articulation of *intrusion detection* as an auditing problem dates to threat-monitoring surveys (Anderson, 1980). Operational systems soon split into *misuse* (signature) and *anomaly* branches (Lippmann et al., 2000). Signatures excel when attack artefacts are stable, but industrial IoT and enterprise backbones now face adversaries who weaponise unknown vulnerabilities—where no string match exists, the defender must infer *normality* and treat strong departures as incidents (Chandola et al., 2009).

This work defends a concrete response to that gap. We do *not* claim to classify attack families; we ask whether a pipeline trained **only on benign** TCP/IP flow statistics can separate normal connections from attacks on a public benchmark. Our narrative follows three ideas from the literature: (1) *support estimation* via one-class kernels (Schölkopf et al., 2001; Tax and Duin, 2004), (2) *denoising* representations that compress routine traffic (Vincent et al., 2008), and (3) *masked prediction* as a self-supervised signal (Devlin et al., 2019), adapted here to *tabular* flow vectors rather than token sequences.

We use NSL-KDD (Tavallaee et al., 2009), a curated revision of the KDD Cup '99 corpus that reduces duplicate flows and sharpens evaluation. We are explicit about limitations stressed

by Mahoney and Chan (2002): traffic is non-stationary, and the NSL train/test split is a *research protocol*, not a guarantee of i.i.d. deployment data. Within that honest scope, we report reproducible metrics and a *failure analysis* of our hybrid stage.

Contributions.

- A semi-supervised pipeline: preprocessor fit on benign rows only; masked autoencoder; one-class SVMs on $\boldsymbol{x}$ and on latent $\boldsymbol{z}$; hybrid score with validation-tuned convex weight.
- Empirical ablation on NSL-KDD (Models A/B/C) with auxiliary Mahalanobis, GMM, and max-$|z|$ baselines.
- Analysis of hybrid under-performance: fusion of non-comparable scores harms ROC ranking despite high precision.

2 Dataset and exploratory analysis

NSL-KDD. Each record aggregates one logical connection: duration, protocol/service/flag symbols, byte counts, rate statistics, and host-level counters (Tavallaee et al., 2009). We use `KDDTrain+_20Percent` for development-speed training and the full `KDDTest+` for reporting, matching common coursework practice. Labels are collapsed to binary: *normal* vs. *attack*.

Patterns that drive preprocessing. Exploratory analysis (correlation heatmaps and kernel density estimates by label) reveals heavy-tailed count-like fields (e.g., `num_root`, `num_compromised`) with extreme skew. Heavy tails inflate variance under Euclidean metrics and destabilise covariance estimates in high dimension after one-hot expansion. We therefore apply $\log(1 + x)$ to the top-k skewed numeric columns *selected on the training split only*, then median imputation and standardisation for numeric columns and dummy encoding for symbolic fields. The `ColumnTransformer` is **fit exclusively on benign training rows** so attack statistics do not leak into location/scale estimates.

Caveats. Attack classes are heterogeneous; treating them as a single "out" distribution simplifies evaluation but obscures per-family behaviour. Temporal correlation across flows is ignored—a known weakness of row-wise benchmarks (Mahoney and Chan, 2002).

3 Methods

3.1 Baselines and advanced classical detectors

Let $\boldsymbol{x}_i \in \mathbb{R}^d$ denote the preprocessed feature vector. On benign training indices $\mathcal{N}$, we compute column means $\boldsymbol{\mu}$ and standard deviations $\boldsymbol{\sigma}$ and score $\max_j |(x_{ij} - \mu_j)/\sigma_j|$ (max-$|z|$). For Mahalanobis distance, we employ Ledoit–Wolf shrinkage of the covariance to stabilise $\boldsymbol{\Sigma}^{-1}$ when d is large (Ledoit and Wolf, 2004; Chandola et al., 2009). For a diagonal-covariance Gaussian mixture with K components, we maximise the expected complete-data log-likelihood via EM (Chandola et al., 2009); K is chosen by BIC on benign data. Anomaly scores are negative log-likelihood. Finally, the ν-SVM one-class formulation (Schölkopf et al., 2001) with RBF kernel defines a smooth decision boundary in a reproducing kernel Hilbert space; we use $\nu = 0.08$ and $\gamma =$ `scale` as a first-pass configuration.

3.2 Masked autoencoder (Model B)

We use a two-hidden-layer MLP encoder–decoder with batch normalisation, dropout, and Xavier initialisation. During training, a random mask $\boldsymbol{m}_i \in \{0, 1\}^d$ with fixed masking probability

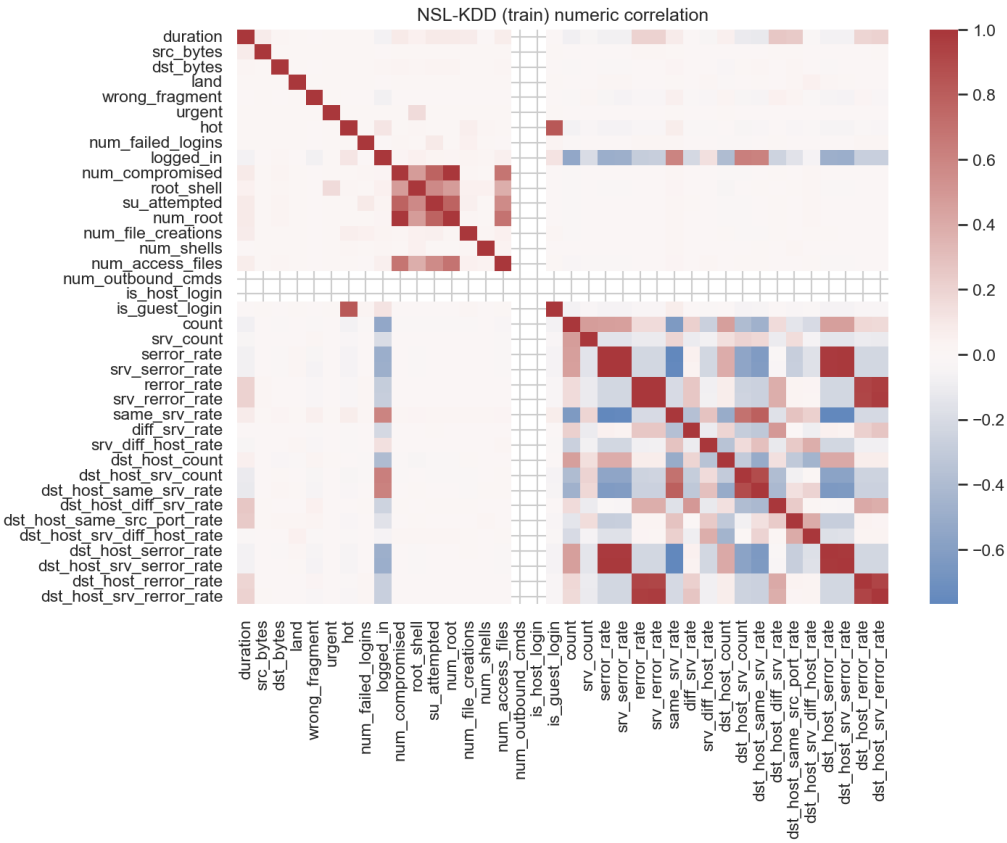


Figure 1: Pearson correlation among numeric NSL-KDD features (training sample). Block structure motivates regularised distance metrics (Section 3).

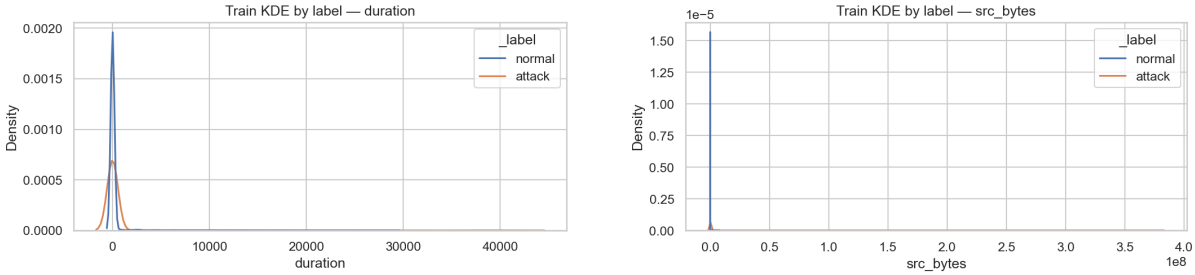
masks coordinates of $\boldsymbol{x}_i$; the network predicts the original vector and minimises mean squared error *only on masked entries*—a direct analogue of masked language modelling (Devlin et al., 2019; Vincent et al., 2008). Optimisation uses AdamW with weight decay; early stopping monitors masked validation loss on benign validation traffic. At test time, reconstruction error is the mean squared error over *all* coordinates for a full forward pass, yielding a scalar anomaly score per flow.

3.3 Latent one-class SVM and hybrid (Model C)

Encoder outputs $\boldsymbol{z}_i \in \mathbb{R}^{d_z}$ are frozen. A second RBF one-class SVM is fit on benign $\boldsymbol{z}$ after per-feature standardisation. Reconstruction scores r_i and SVM margin scores s_i are min–max normalised using statistics from *benign validation* rows only, then combined as $h_i = wr_i + (1-w)s_i$ with w selected on a coarse grid to maximise validation F1 at an F1-optimal threshold.

3.4 Training protocol and metrics

We hold out a stratified 15% validation slice of the training file for threshold tuning. For each model, we choose the threshold maximising F1 on validation, then report ROC-AUC, PR-AUC (average precision), precision, recall, and F1 on `KDDTest+`. ROC-AUC summarises ranking quality independent of the chosen threshold.



Duration by label

Source bytes by label

Figure 2: Label-conditional KDEs for selected heavy-tailed fields (after optional $\log(1+x)$ in pipeline). Separation visible on duration and byte volume motivates skew-aware transforms.

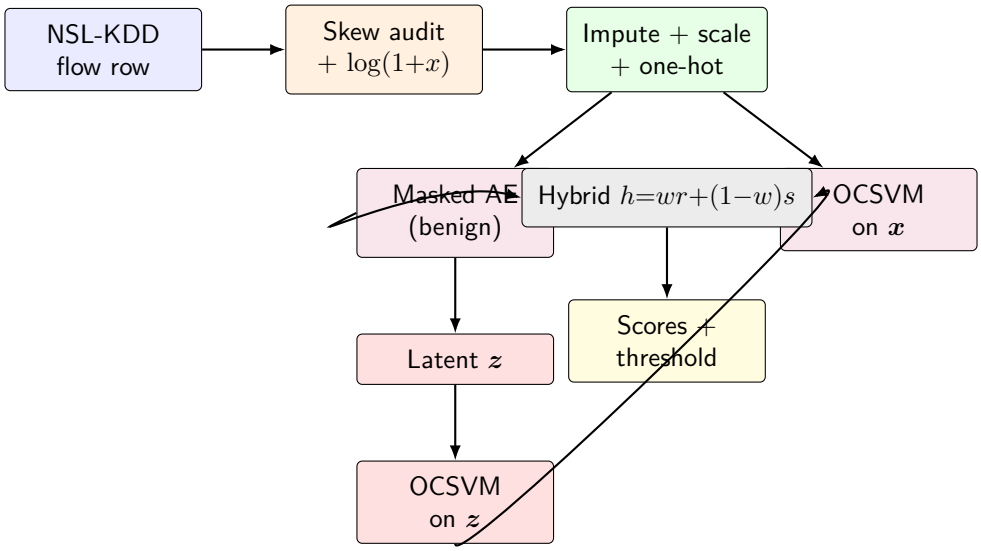


Figure 3: End-to-end processing graph. Preprocessor and all detectors are fit on benign training data only; validation selects thresholds and hybrid weight.

4 Results

table 1 summarises Models A–C on `KDDTest+` using thresholds tuned for F1 on validation. fig. 4 visualises ROC-AUC across main and auxiliary models.

Reading the tables. Model B achieves the highest ROC-AUC among A/B/C, indicating that reconstruction error ranks normal vs. attack flows slightly better than the raw-feature SVM under our preprocessing. Model A remains competitive and offers a lighter runtime profile (no neural training). Auxiliary Mahalanobis achieves the peak ROC-AUC in our suite, highlighting that elliptical distance after shrinkage is remarkably effective on this feature space—a useful sanity check for practitioners.

5 Failure analysis and discussion

Hybrid degradation (Model C). ROC-AUC collapses from ≈ 0.94 to 0.78 after convex fusion. *Why?* ROC-AUC depends on the ordering of scores; min–max normalisation using only benign validation quantiles does not guarantee that reconstructed error and SVM margin are on a commensurate *rank scale*. If one score preserves ordering of hard negatives better than the other, a linear mix can introduce pairwise inversions—destroying AUC while still allowing a

Table 1: NSL-KDD `KDDTest+` results after validation threshold tuning (executed pipeline, `KDDTrain+_20Percent`).

Model	ROC-AUC	PR-AUC	F1	Precision	Recall
A: OCSVM on $\boldsymbol{x}$	0.920	0.942	0.846	0.929	0.777
B: MAE recon. MSE	0.942	0.944	0.832	0.923	0.757
C: Hybrid	0.781	0.879	0.812	0.934	0.719

Table 2: Auxiliary classical scores (same protocol).

Method	ROC-AUC	F1
Max-$\lvert z \rvert$	0.928	0.894
Mahalanobis (Ledoit–Wolf)	**0.950**	0.864
GMM NLL (K=7 by BIC)	0.923	0.803

high-precision operating point (precision 0.93 but recall 0.72). Formally, let $s_i^{(1)}, s_i^{(2)}$ be monotone transforms of latent margins; unless there exists a single w such that $ws^{(1)} + (1-w)s^{(2)}$ is everywhere rank-consistent with the better score, AUC need not improve.

Mitigations. (i) Learn w or a small MLP on validation with AUC as a surrogate; (ii) replace linear fusion with *max* or *logical AND* at calibrated quantiles; (iii) use only latent SVM if reconstruction is redundant; (iv) copula or rank-based normalisation on the full validation distribution including attacks (careful with leakage).

Masked autoencoder training. Early stopping selected weights near epoch 6 in our run—either sufficient capacity for the task or premature plateau; reporting learning curves and multiple seeds would strengthen claims.

Deployment path. Offline NSL rows abstract PCAP processing; live systems would aggregate packets (e.g., via Scapy) into the same statistical schema before scoring.

6 Conclusion

We presented a semi-supervised anomaly detection stack for zero-day-style threats, combining skew-aware feature engineering, masked autoencoder reconstruction, and kernelised one-class machines on NSL-KDD. Strong ROC-AUC (≥ 0.92) for Models A and B supports the thesis that benign-only training can flag unseen attacks in flow feature space. The hybrid's AUC drop documents that *integration is non-trivial*: principled fusion remains open work. Future directions include full-train NSL experiments, hyperparameter grids for ν and γ, sequence encoders over raw bytes, and drift-aware monitoring under non-stationarity (Mahoney and Chan, 2002).

Reproducibility. Code and notebook: repository `dl-aml-cybersec-project`; run `notebooks/phase1_eval_` (or `jupyter nbconvert -execute` on it) and compile this report per `report/README.md`.

Acknowledgements

This report was prepared as coursework at Rishihood University. We thank the open NSL-KDD mirroring community for hosting reproducible copies of the dataset.

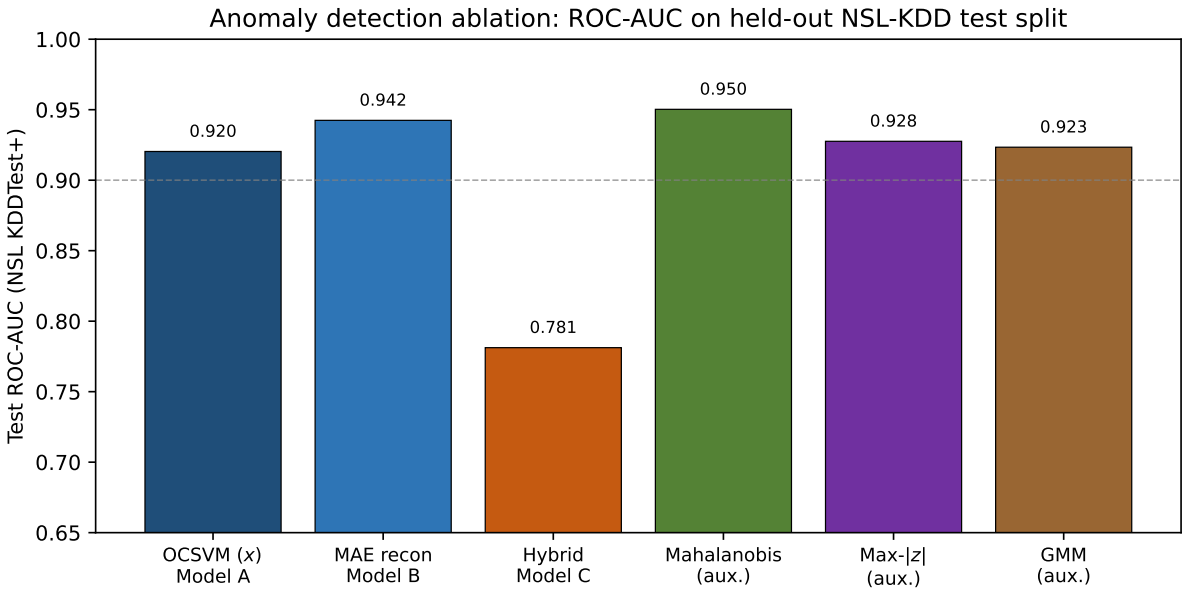


Figure 4: Test ROC-AUC comparison (values from executed notebook outputs). The hybrid underperforms single-signal models on ranking quality.

References

James P. Anderson. Computer security threat monitoring and surveillance. *James P. Anderson Co. Technical Report*, 1980.

Varun Chandola, Arindam Banerjee, and Vipin Kumar. Anomaly detection: A survey. *ACM Computing Surveys*, 41(3):15:1–15:58, 2009. doi: 10.1145/1541880.1541882.

Jacob Devlin, Ming-Wei Chang, Kenton Lee, and Kristina Toutanova. BERT: Pre-training of deep bidirectional transformers for language understanding. In *Proceedings of NAACL-HLT*, 2019.

Olivier Ledoit and Michael Wolf. A well-conditioned estimator for large-dimensional covariance matrices. *Journal of Multivariate Analysis*, 88(2):411–431, 2004.

Richard P. Lippmann et al. Progress in Detecting Distributed Denial of Service Attacks. Technical report, MIT Lincoln Laboratory, 2000.

Matthew V. Mahoney and Philip K. Chan. Learning nonstationary models of normal network traffic for detecting novel attacks. In *Proceedings of the 5th International Symposium on Recent Advances in Intrusion Detection (RAID)*, pages 132–147, 2002.

Bernhard Schölkopf, John C. Platt, John Shawe-Taylor, Alex J. Smola, and Robert C. Williamson. Estimating the support of a high-dimensional distribution. *Neural Computation*, 13(7):1443–1471, 2001.

Mahbod Tavallaee, Ebrahim Bagheri, Wei Lu, and Ali A. Ghorbani. A detailed analysis of the KDD CUP 99 data set. In *Proceedings of the IEEE Symposium on Computational Intelligence for Security and Defense Applications*, 2009.

David M. J. Tax and Robert P. W. Duin. Support vector data description. *Machine Learning*, 54:45–66, 2004.

Pascal Vincent, Hugo Larochelle, Yoshua Bengio, and Pierre-Antoine Manzagol. Extracting and composing robust features with denoising autoencoders. In *Proceedings of ICML*, 2008.